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 Studierichting: Informatica
 Oefeningenreeks 1C nr. 31.8

$$\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{6}}\right)^8$$

$$r = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{6}}\right)^2} = \sqrt{\frac{2}{3}}$$

$$\sin \theta = \frac{\frac{1}{\sqrt{6}}}{\sqrt{\frac{2}{3}}} = \frac{1}{2}$$

$$\cos \theta = \frac{\frac{1}{\sqrt{2}}}{\sqrt{\frac{2}{3}}} = \sqrt{\frac{3}{4}}$$

Eerste kwadrant:

$$\theta = \frac{\pi}{6}$$

$$z = \left(\sqrt{\frac{2}{3}}\right)^8 \left(\cos\left(\frac{8\pi}{6}\right) + i \sin\left(\frac{8\pi}{6}\right)\right)$$

$$z = \frac{8}{81} (-1 - \sqrt{3}i)$$

$$z = -\frac{8 + 8\sqrt{3}i}{81}$$

References